

Day 2 — Two-Digit $\times$ Two-Digit Near 100 (Below)

Module: Multiplication Near a Power of 10 — Nikhilam · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: compute 97×96 using deficits.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *vinkulam* (IAST: vinkulam; lit. “deficit (number below base)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. The two-step rule for both factors below 100.

- Find deficits from 100: e.g., $97 \rightarrow -3$, $96 \rightarrow -4$.
- **Left part:** cross-subtract. $97 - 4 = 93$ (or equivalently $96 - 3 = 93$).
- **Right part:** product of deficits. $3 \times 4 = 12$.
- Combine: **9312**.

2. Worked examples.

- $97 \times 96 = 93 \mid 12 = \mathbf{9312}$.
- $88 \times 95 = 83 \mid 60 = \mathbf{8360}$. (Deficits: 12, 5. Cross: $88 - 5 = 83$ or $95 - 12 = 83$. Product: $12 \times 5 = 60$.)
- $99 \times 99 = 98 \mid 01 = \mathbf{9801}$. (Deficits 1, 1. Cross: 98. Product: 1. Pad to 2 digits: 01.)

3. Why the padding matters.

The right part must have as many digits as the base has zeros (here, 2 zeros $\rightarrow$ 2-digit right part). If product is single-digit, pad with leading zero. If product overflows (3+ digits), carry to left part.

4. Algebra.

Base $B = 100$. Factor 1 = $B - a$, factor 2 = $B - b$. Then $(B - a)(B - b) = B(B - a - b) + ab = B \cdot (\text{cross}) + (\text{right})$. The cross-subtract IS $B - a - b$. The product of deficits IS ab .

5. Independent practice.

8 problems, all below-100 case. 10 minutes.

Activity (15 min)

A scaled version of *Method Speed Race* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Two-Digit $\times$ Two-Digit Near 100 (Below).

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).